# What we have

Every finding of the theory on 29 October 2026, with the corrections of the council and the debate

Read together, the findings show one pattern. When a learning attack has a price, either the price binds every estimator, or it comes from a restricted class of attackers or from hard structure placed in the record law. No result prices a learning attacker as a learner, a shadow model or a VAE, with a lower bound that sees its class. The oracle and measurement layer is settled. The statistical prices are many and exact. Every computational price maps the attack onto statistical queries, low-degree methods or a lattice problem, and each needs planted structure in the record law. The debate gave the first price with a measured side and a proven side, in a setting that turned out to be the Gaussian single-index model. The booklet holds 63 objects: 30 accepted, 29 borrowed, 2 open and 2 refuted. Milestone 2 added eight results to the registers, the council five results that both critics accepted, and the debate ten objects. None is checked by you. Section 10 merges the eighteen decisions of five earlier readings into nine.

Every result below is proven and accepted unless its label says otherwise. Booklet numbers refer to `lib/theory/booklet.pdf`. "Milestone 2" marks a result in `sink/formalizing/results.md` that is not yet in the booklet. "Council" marks a claim of `lab/theory/council/2026-10-27-theory-directions/`, with the two critics' verdicts. No council claim is accepted in the sense of `CONTEXT.md`, since the council was not a round of a space. "Corrected" marks a result whose statement the council or the debate narrowed.

## 1. The objects

The *learning oracle* serves a learning function $\mathsf{L}(x; \rho, \xi)$, which maps a training input $x$, a seed $\rho$ and hardware randomness $\xi$ to a model. It answers $\mathsf{Train}(x, s)$, Eval and Open. A caller in the named mode may reuse a seed, and one in the drawn mode may not. The bottom oracle withholds the honest seeds and the top oracle hands them over. The *twin* of a run is the same run, with the same seed, on the neighbouring dataset. The coupled tail is the probability that a run and its twin differ by more than $\Delta$. A *certifier* outputs an upper bound on a leakage target from Train calls, and it is sound when the bound holds for every learning function. The definitions reading of 27 October states these objects in full, as Definitions 1 to 8.

A price is a set of minimal budgets. The definitions reading proposes one form for every attacker class $\mathcal{A}$. The price $\mathrm{price}_{\mathcal{A}}(s)$ holds the minimal budgets at which one member of $\mathcal{A}$ reaches success $s$ on every instance of a class. Three attacker classes have games: the unbounded class, with reference samples and Train calls as budget, the statistical-query class, with queries and a tolerance, and the polynomial-time class under a named assumption. The debate added the *statistic class* $\mathcal{A}_S$, the attackers that read the observations only through a statistic $S$, such as their empirical covariance. A learning attacker, meaning a learner of a declared architecture and optimizer with a budget in learning units, has no definition.

Two record laws recur. The *pancake law* is Gaussian except along a hidden direction $v$, where it concentrates on thin parallel layers. Detecting it is homogeneous CLWE. A *single-index law* has labels that depend on $Z = \langle u, x \rangle$ alone. Its *generative exponent* $k^*$ is the least Hermite degree $k$ with $\mathbb{E}[h_k(Z) \mid Y] \neq 0$, and its information exponent is the least $k$ with $\mathbb{E}[Y h_k(Z)] \neq 0$. The generative exponent never exceeds the information exponent (DPLB24, Definition 2.4).

## 2. The oracle and measurement

1. The oracle is realizable (`thm-perfect-substitution`). The real procedure, run with fresh randomness, gives the bottom oracle's joint law of honest values and transcript. Every bound proven under the bottom oracle holds for the real system.
2. An unbounded attacker gains nothing from Train calls (booklet 6.4). Its profile is flat in runs, so a profile is measurable only for a named family of attacks. The oracle has value only to a bounded attacker, and no result prices that value.
3. A certifier needs the seed name (booklet 3.8, 6.4). For one training run, the leakage of any channel is at most the coupled divergence over same-seed pairs. Named-seed runs certify it for every learning function. Drawn-seed runs cannot certify a total variation below one through a deterministic channel.
4. Needle lower bound (booklet 3.6). If $J$ learning functions differ from a reference only on disjoint inputs, a certifier with $N$ Train calls is fooled at the reference with probability at least $1 - \alpha - N/J$.
5. Black-box certifiability (booklet 3.9). With finitely many queries and outputs, drawn-seed runs certify the best advantage of $q$ label queries, with error falling as the inverse square root of the number of runs.
6. Composition and deviation (booklet 3.10, 3.11). A client isolated behind Gaussian releases leaks at most a Gaussian term plus the coupled tail, and coupled runs certify that tail with a number of runs free of the dimension. A coalition that deviates by at most $B_A$ raises the tail to a fixed function of the honest one, and certifying it costs about $e^{\mathfrak{e}}$ runs with $\mathfrak{e} = (R-1)B_A^2/(2\sigma^2)$, matched by a lower bound.
7. Drawn seeds stay sound (booklet 6.1 to 6.3). Every certificate above stays sound under independent drawn runs for bottom-oracle targets. A shared seed improves accuracy and leaves soundness unchanged.
8. The price of twins (booklet 6.5). A sound certifier that shows near-zero leakage makes about $\ln(1/\alpha)/\epsilon_{\mathrm{acc}}$ calls at the twin input, and runs at any other input count for nothing.
9. Replay-audit steering (booklet 3.12). With seeds from a random oracle, a prover with $q$ queries passes $n$ replays with a chosen model at most $(q+1)n$ times as often as an honest run lands there. It uses the random oracle as a plug-in. The council found it to be a Fiat-Shamir audit. It fails under every function used in place of the random oracle, over the class of all learning functions. An interactive audit, in which the auditor draws the seed after the commitment, removes the plug-in (`prop-replay-audit-uninstantiable`, fable REVISE, sol ACCEPT).
10. The oracle read as a random oracle (council). Four of the five random-oracle techniques already have instances in the theory. Admissible programming changes no price (`prop-programming-preserves-prices`, REVISE by both).

## 3. Statistical prices

11. Shadow-run price (booklet 4.3). Coming within $a$ of the best advantage on every learning function with $k$ possible views costs $\Theta(k/a^2)$ Train calls. Chosen or opened seeds do not lower it.

12. Label sandwich (booklet 4.4). From a representation $Z$, an unbounded reconstructor succeeds at least as often as the class baseline minus the label error, and at most that baseline plus $\sqrt{I(X;Z \mid Y)/2}$.
13. Orbit limit and price (milestone 2, corrected). A seed-blind server learns the records only up to the symmetry group of the record law. For Gaussian records it cannot place one record within $\sqrt{d}/(2\sqrt{2})$ except with probability $e^{-\Omega(d)}$. For linear clients the minimax price up to rotation is $\Theta(d^2/N)$ per record. The council narrowed it to SGD, since Adam is equivariant only under signed permutations, which have invariants of degree 4 beyond the Gram matrix.
14. ReLU clients give up their weights (milestone 2). For Gaussian records, the second moment of a Gaussian ReLU client's outputs fixes its weights up to rotation, so the server inverts every record up to that rotation in polynomial time. This holds at the population level, and the sample price is open.
15. Same-units gap (milestone 2). For the Gaussian mechanism on the mean with noise scale $\rho = n^{-a}$, $0 < a < \frac{1}{2}$, membership costs the attacker $\Theta(1 + n\rho^2)$ times the trainer's price, and an efficient attack attains it. The hypothesis is that the noise follows the population covariance.
16. Bayes release (council, `thm-bayes-release-loss-matched-price`, fable ACCEPT, sol REVISE for integrability). A Bayes release answers every correlational query exactly as the data do. A square-loss gradient learner therefore pays the same to extract it as to train on the data.
17. Targeted reader (council, `prop-targeted-reader-no-learning-price`, both ACCEPT). An attack on one candidate record faces two laws. Only statistical bounds apply to it, and it has no learning price.
18. The trainer as a filter (council). Noisy population gradient descent sees a family of laws only through the mean shift of its per-example update. The trajectory's total variation from the null is at most $\sqrt{T\bar{\kappa}_f/(4\varsigma^2)}$, and this bounds every map of the trajectory, white box included. For $k$-sparse parities it forces $T \geq 4\varsigma^2\binom{n}{k}/(9G^2)$ (`thm-noisy-gradient-query-filter`, both ACCEPT). The slogan that what the trainer did not learn stays hidden is false. A trainer that outputs one bit, or an $\varepsilon$-differentially private bit, hands over a planted parity (`prop-output-bounds-do-not-filter`, fable ACCEPT, sol REVISE). A filter must restrict the trainer while it trains. Under Gaussian noise the attacker's advantage has a population term and a term for the information the trajectory holds about the sample (`thm-noisy-trainer-two-term-filter`, fable ACCEPT, sol REVISE). A hashed-table trainer shows that the second term is needed (`prop-hashed-memory-breaks-sq-filter`, both ACCEPT). fable reads the composed bound as differential-privacy accounting with Fano's inequality, vacuous for every deployed run.
19. Pseudorandomness for a learner class (council, REVISE by both). The notion "a class cannot tell the joint law from the product of its marginals" is outcome indistinguishability of the constant predictor. In each class it reduces to a known quantity: total variation, statistical dimension, or the generative exponent. It equals learning in log loss only.

## 4. Computational prices

20. The seed separates attackers (booklet 5.6 to 5.8, corrected). On pancake records behind a Gaussian ReLU client, a seed-known server inverts every record with one linear program each. A seed-blind statistical-query server needs $\exp(\Omega(c_0^4 d))$ queries for correlation $c_0$. A seed-blind polynomial-time server reconstructs finely only if homogeneous CLWE is easy. The council narrowed it in two ways. It holds for SGD only, as in item 13. A real seed is also a short key. For seeds whose representation laws are separated, about 400 representations identify a 64-bit seed,

so every seed-blind lower bound is capped at the number of seeds (`thm-short-seed-dichotomy`, fable ACCEPT and recomputed as 397.3, sol REVISE). Seeds with identical representation laws, such as a client and its negation, stay hidden.

21. Coarse reconstruction is lattice-hard (milestone 2). Reconstructing pancake records to any constant correlation is as hard as homogeneous CLWE, since two independently rotated reconstructions that agree reveal the hidden direction. Proven in exact arithmetic.
22. The two attacker games differ (milestone 2). With exactly discrete hidden labels behind a hidden linear client, a polynomial-time server reads the labels from $d + 1$ representations by lattice reduction, while every statistical-query tester needs $2^{d^{\Omega(1)}}$ queries or polynomially small tolerance. Proven in exact arithmetic.
23. Quantized records and ZLIP (milestone 2, sketch). For records from the integer lattice, blind reconstruction is at least as hard as recognizing a rotated $\mathbb{Z}^d$, and for a linear client no harder. One blocking challenge on the arithmetic model is open.
24. A good predictor gives up the label direction (booklet 5.9, 5.10). A server holding the top model with excess risk $\varepsilon$ recovers the label direction within $O(\sqrt{\varepsilon})$ and reconstructs that coordinate of every record. From the labels alone, that coordinate costs $d^{k^*/2}$.
25. Extraction against a squared-label trainer (milestone 2, corrected). With labels $\mathbb{E}[Y \mid Z] = |Z|e^{-Z^2/2}$, the trainer finds the direction from $O(d)$ records, while a statistical-query extractor of the release from random inputs needs VSTAT($c\,d^2$). The council removed its reading as a learning one-way function. Against a square-loss gradient trainer the gap vanishes (item 16). It exists only against a trainer that reads squared labels, and the information exponent of the label law bounds it (`prop-extraction-gap-information-exponent`, fable ACCEPT, sol REVISE). The degree-4 link is a knife edge. Every even link monotone on the half line has generative exponent at most 2, because each level set is a pair $\{\pm a\}$ on which the second Hermite moment is nonzero (fable, checked by the integrator).
26. A learner's SQ bound binds every attacker (booklet 5.2, generalized in milestone 2, corrected). If a procedure reads its data only through one-bit statistical queries, any SQ-hard test bounds every attacker of its output. The SGD clause holds only in the worst-rounding model of AKMSS21, in which memorization cannot be stated (`thm-null-trajectory-filter` (b), both ACCEPT).
27. The degree of a channel (council, `thm-shadow-price-is-generative-exponent`, both ACCEPT). It is the generative exponent of DPLB24 with the secret as the label, and the shadow-learning price is their lower bound. The seed cannot grade the price. Drawn seeds give one mixture law and a named seed gives degree 1 (`prop-seed-cannot-index-the-family`, REVISE by both). Every degree above 2 comes from the moments of the record law along a hidden direction, so the frame moves the planted structure into the record law.
28. The exponent depends on the frame (council, `prop-overlap-grading-carries-price`, fable ACCEPT, sol REVISE). At one degree $k$ the price is $d^{k/2}$ when the hidden index is a Gaussian direction and $d^k$ when it is a subset.

## 5. The debate of 28 and 29 October, a learning attacker priced

The setting (`def-optimum-update-stream`). Inputs are Gaussian, and labels are $Y = \xi W$ with $\xi$ an independent sign and $W$ a function of $Z$. The labels have conditional mean zero, so the trained linear model is the same for every secret $u$. In one FedSGD round at that optimum, the server sees $T$ client updates.

29. Every observer (`thm-update-direction-chi-square-bound`,

`prop-update-direction-fano-recovery`). Recovering $u$ to overlap $\frac{1}{2}$ with probability $\frac{2}{3}$ needs $T \geq (d/24 - 0.93)/\kappa_{\mathrm{KL}}$ updates, with $\kappa_{\mathrm{KL}}$ the Kullback divergence of one update along $u$.

30. The weighted spectral estimator (`thm-norm-weighted-update-pca`, `prop-signed-weight-update-pca`). Weighting each update by a function of its norm before PCA gives $M_g = \alpha_g I_d + \gamma_g uu^\top$, and the top or bottom eigenvector finds $u$ after a number of updates set by the spike and the condition numbers. Plain PCA is the constant weight.
31. Measured meets proven (`prop-measured-meets-proven-full-observer`). On the challenger's law $Q_0$, $\kappa_{\mathrm{KL}} \leq 0.1602$ from a quadrature value with an error estimate (numerical). Every estimator then needs at least 46.25 updates at $d = 200$ and 98.27 at $d = 400$. The weighted estimator reached overlap $\frac{1}{2}$ in 40 of 40 declared runs at $T = 20d$ (empirical). The price of the full observer is known within a factor 86.5 at $d = 200$ and 81.4 at $d = 400$.
32. The linear class (`def-statistic-learner-class`, `conj-linear-class-reads-cauchy-transform`, open). PCA and a linear VAE read only the empirical covariance. The conjecture says this class reads a Cauchy-type transform of one signed measure of the label law, while the full observer reads a weighted $L^2$ norm of the same measure. Its finite identities are proven, two deterministic equivalents are heuristic, and the outlier equation matches simulation within 0.02 (empirical). On laws with $c_2 = 0$, PCA stays at or below overlap 0.09 while the weighted estimator reaches 0.86 at $T = 40d$, $d = 200$ (empirical).
33. A defence (`prop-full-clipping-optimum-filter`). Per-example normalization of the gradients at a median-fitting optimum leaves each update only the sign of its residual, and closes the channel exactly for every observer. Clipping below $w_{\min}\sqrt{d}/2$ equals normalization except with probability $Te^{-9d/64}$ (proven, given the Laurent and Massart tail bound, from memory).
34. Re-learning a release (`thm-release-relearning-price-bracket`). In the Gaussian linear model, an attacker that queries a trained release needs at least $(n - d - 1)/\kappa$ and at most $d + 1 + (n + s^2/\tau^2)/\kappa$ answers to come within a factor $1 + \kappa$ of the trainer's risk, when the release answers with nature's noise at inputs no larger than nature's. Least squares attains it. Without that noise or that bound on query size, $d$ queries suffice. Your sentence on "the performance of initial model" has three readings, and all three are now settled. The trainer's performance caps the attacker's accuracy, floors its success, and floors its cost only under that noise and that bound on query size.

The critics. fable showed that at batch one each update is a labelled sample of a single-index model with the secret as hidden direction, so the full observer's price is the degree-2 price of DPLB24 (Theorems 3.2, 4.1, 6.1), and the gap between PCA and the weighted estimator is their Proposition 2.6. The outlier equation and the optimal weight belong to spectral initialization (Lu and Li 2020, Mondelli and Montanari 2019, Luo, Alghamdi and Lu 2019, from memory, none in the library). The laws on which PCA fails satisfy $c_2 = 0$ by construction, a balanced moment. On a generic law the linear class is within a constant factor of optimal. Hardness is impossible in this setting, because it has degree 2. sol found the release theorem to be Gaussian conjugate regression with an A-optimal-design trace bound, and asked six local repairs. fable accepted nine objects and asked one revision, sol accepted four and asked six, and neither found a false statement.

## 6. The computational price of a learning problem, from the literature

This section is the integrator's synthesis of 29 October, from the library notes and, where marked, from memory. For a Gaussian single-index problem in dimension $d$, each attacker class has its own price.

- Every estimator needs of order $d$ samples (proven).

- Statistical-query and low-degree learners need $d^{k^*/2}$, up to the factor $\lambda_{k^*}^{-2}$ (DPLB24 Theorem 3.2, proven for SQ, Theorem 3.5 under the low-degree conjecture). A partial-trace algorithm attains it (Theorem 4.1).
- Correlational learners, which use the label without transforming it, pay $d^{l^*/2}$ with $l^*$ the information exponent (from memory, unverified). Full-batch gradient descent on the square loss is such a learner.
- Online SGD on one neuron from a random start pays $d^{l^*-1}$ for $l^* \geq 3$ (Ben Arous, Gheissari and Jagannath, JMLR 2021, from memory, unverified, not in the library).

The computational aspect of the problem is the distance between the first price and the others. The last two bind gradient learners of a fixed form, so the claim of the council and the debate that no tool binds a gradient learner is too broad. The true claim is narrower. No tool binds a gradient learner on a law outside the single-index family. Outside it, the statistical dimension of Feldman et al. (J. ACM 2017) gives the statistical-query price of any problem posed on a specified law. AKMSS21 Theorem 1 says when that price binds mini-batch SGD on a differentiable model with batch $b$ and gradient precision $\rho$. With $b\rho^2$ large, SGD is no stronger than statistical queries. With $b\rho < 1/8$ it simulates any sample-based learner (proven, with an adversarial last rounding bit). A real architecture on real data has no law, and only the measured cost of a declared learner remains.

Hardness against every efficient learner on a samplable law needs an assumption. Blum, Furst, Kearns and Lipton (CRYPTO 1993) build a one-way function from any learning problem that is hard on average (from memory, unverified, not in the library). Hardness therefore has two sources. A restricted class gives an unconditional bound that white-box algorithms escape, as Gaussian elimination escapes the SQ bound for parity. A conjecture gives a bound against every efficient algorithm. A natural learning problem that yields a one-way function, which `CONTEXT.md` counts, is of the second kind.

## 7. What was refuted or withdrawn

- Our own. The proof note's black-box reduction (booklet 5.1) and the conjecture that reconstruction costs what training costs (booklet 5.11) are refuted. The debate withdrew five objects. Four were superseded by objects of items 29 to 34, and the conjecture of a degree gap in the update moments was withdrawn outright. The round-03 summary of the space names each and records its fate.
- Earlier readings. "An efficient learner outputs a cheap predictor" holds for circuits and fails for uniform descriptions (fable). The three-party frame of the reading of 27 October does not survive as a theory, since its single-party notion is item 19 and its degree is item 27 (council).
- The literature. FORMAL_SECURITY's `.tools/proof.tex`, main theorem false by a half-page counterexample (booklet 5.1). Goldwasser, Kim, Vaikuntanathan and Zamir, Appendix A, whose sparse-PCA backdoor is detected at its printed parameters, while their main CLWE construction stands. Feldman et al., J. ACM 2017, Theorem 3.16, one wrong sign. Mukherjee and Lokam, arXiv 2303.16372, Theorem 10 and Corollary 12, not established. Zadik, Song, Wein and Bruna, COLT 2022, Theorem 3.3, which fails when all labels but the first vanish.

## 8. What is open

- The seed is worth exactly the NGCA price of the record law (booklet 5.12). Settled from below, and from above for linear clients.
- A same-units gap for low-rank adapters (booklet 7.1).

- `conj-linear-class-reads-cauchy-transform` (item 32), and the recovery threshold of the update observer at batch $b$ and noise $\varsigma$, near $T \asymp d(1 + b\varsigma^2)^2$ (council, heuristic).
- The sample price of blind ReLU inversion (item 14), extraction of a fitted rather than exact release, finite-precision versions of items 21 to 23, and the blocking challenge on item 23.
- A natural-data oracle, parked by you on 27 October (`prob-natural-data-oracle`).
- The core. A lower bound for a gradient learner of a fixed architecture on a law outside the single-index family. No result of ours and no tool in the library reaches it.

## 9. What it adds up to

The oracle and measurement layer is proven for every learning function, and only item 9 changed after it was written. The statistical side prices shadow runs, reconstruction, membership and update observation exactly, and the debate brought one of these prices within a stated factor by experiment. The computational side has one mechanism. A hidden direction or subset in the record law carries the hardness, and the attacker class is SQ, low-degree, a statistic, or polynomial time under a lattice assumption. Each time the council or the debate took the simplest channel that training produces, it reduced to a known single-index or regression problem. Items 2 and 26 locate the open core. The oracle has value only to a bounded attacker, and the only bound that sees a trainer's class holds in a rounding model where memorization cannot be stated.

## 10. Decisions

The eighteen decisions of the readings of 26, 27 and 29 October reduce to nine. Each names the decisions it replaces.

1. The booklet. Freeze Definitions 1 to 8, and merge milestone 2, the council's corrections and the debate's objects, with attributions to the known results. Replaces 26 October decision 6, definitions decision 1, debate decision 3.
2. The searches. Run them before anything is claimed new: outcome indistinguishability, multicalibration and pseudoentropy, and spectral initialization with the exact limits of generalized linear models. AS20 and ABM23 sit in the library unread. Replaces aim decision 3, debate decision 1.
3. The attacker class. Adopt the gradient attacker, embedded in the statistical-query class through AKMSS21, knowing that the class to end with is the open core. Or let each statement name its class, as section 6 does. Replaces definitions decision 2, aim decision 2.
4. The plug-in boundary. Does the carrier rule of the mapping definition read "a cryptographic plug-in does not count" as you meant it? Does a pseudorandom seed generator, needed by item 20 on real implementations, count as the excluded plug-in? Replaces 26 October decision 4, council ruling 1.
5. The programming rule. Your words of 27 October, "indistinguishable from random wheter it has returned a pre fixed data point", read narrowly. Under that reading, programming makes canaries, recorded seeds and known noise into measurement devices, and it changes no price. Confirm the narrow reading, or adopt the broad one, under which release noise counts as a defence. Replaces council ruling 2.
6. Canaries. Honest draws measure the average leakage of a typical record, and out-of-distribution records measure the worst case. Replaces council ruling 3.
7. Exact arithmetic. Are items 21 to 23 acceptable in exact arithmetic for now, and should a round go to the ZLIP challenge? Replaces 26 October decisions 2 and 5.

8. The Train-access model of item 15. Replaces 26 October decision 1.
9. The next statement. fable proposes that batching makes the linear attacker optimal, with first correction $\mathrm{Var}(W^2 T(W))/(b\, c_2^2)$. sol proposes pricing one-pass linear SGD and charging the attacker's optimizer work. The alternative is to stop the line. Replaces debate decision 2.

Three earlier decisions are answered and need nothing from you unless you overrule. The three-party frame (aim decision 1) did not survive the council. The exponent alone is no unit (definitions decision 3), by item 28. Extraction from random inputs (26 October decision 3) has no gap against square-loss gradient trainers, by item 25.